# Where does TR4's rooftop PV go?

Prosumers, pure customers, and export — energy accounting of one representative day

Feeder TR4 · rolling-horizon exact-QCP OPF · 48 half-hours · 2026-07-24

## Context

Transformer TR4 serves 75 connections: 46 prosumers (rooftop PV and a battery, operated for self-consumption) and 29 pure customers (load only, no PV or battery). The prosumers generate 1,163 kWh of PV over the day — about 3.4× their own demand (343 kWh). This note tracks every kWh of that PV to its destination.

> Key finding. The rooftop PV does not serve only its owners. 100.8 kWh/day — 8.7 % of all PV — flows to the 29 pure customers, covering 49.7 % — just under half — of their demand. The other 50.3 % is imported from the grid at night, when there is no PV. Roughly half of the customers' electricity therefore comes from their neighbours' rooftops.

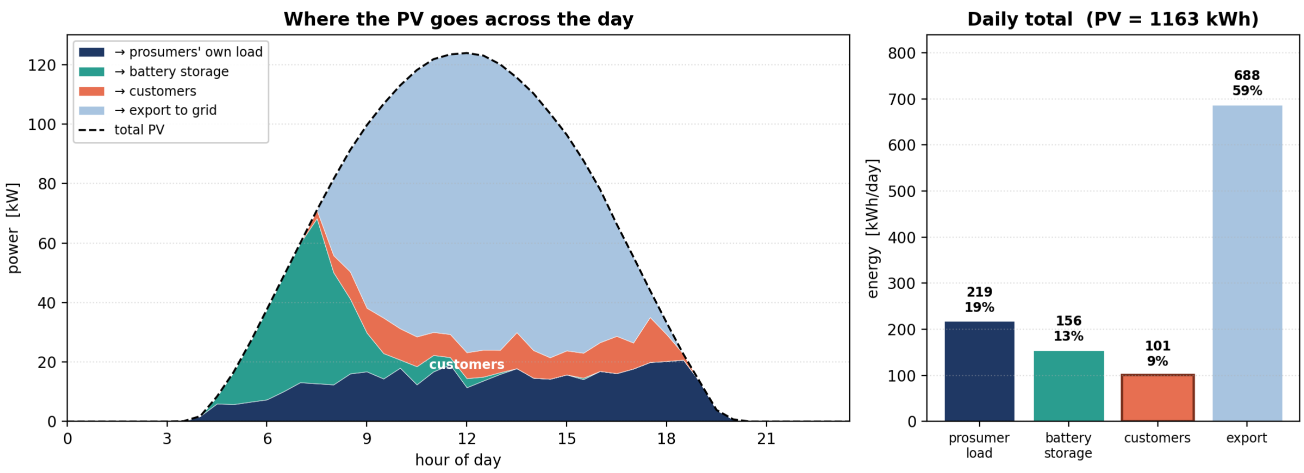


Figure 1: Left: the PV split half-hour by half-hour. In the early morning the batteries absorb the surplus; once they fill up (mid-morning) the surplus goes to customers and export. Right: daily totals.

## Where the day's PV goes

| Destination | kWh/day | % of PV |
|---|---|---|
| Prosumers' own load (direct) | 218.5 | 18.8 % |
| Battery storage | 155.8 | 13.4 % |
| Customers | 100.8 | 8.7 % |
| Export to grid | 688.0 | 59.2 % |
| Total (= PV) | 1,163.0 | 100 % |

Table 1: PV disposition (priority allocation).

| Customer supply | kWh/day | % |
|---|---|---|
| From prosumer PV (day) | 100.8 | 49.7 % |
| From grid import (night) | 102.0 | 50.3 % |
| Total customer load | 202.7 | 100 % |

Table 2: How the 29 customers are supplied.

## Reading the day

At the noon peak the PV reaches 124 kW. Only 9 % serves prosumer load; the batteries are nearly full and absorb almost nothing (2 %); 7 % serves the customers; the remaining 81 % is exported. Early in the morning the batteries take the surplus first — up to 56 kW charging around 07:30. Once they saturate in mid-morning, the surplus has two outlets left: the customers and the grid. The battery is the only controllable lever, and it stops helping once full.

## Method and sources

Priority energy allocation, applied every half-hour: PV serves the prosumer's own load first, then charges the battery, then serves customer load, and exports the rest. Grid import covers any load the PV cannot. Downstream network and battery round-trip losses (about 3–4 % of PV) are not separated out here. Inputs: per-connection load and PV profiles (`TR4_load_PV_48periods.xlsx`, from OpData); battery dispatch (exact-QCP rolling OPF, `RunFOR_Rolling`, CPLEX); classification by PV size (sheet `Customers`). Every figure in this note is reproduced by live formulas in the companion workbook **`TR4_PV_allocation_prosumers_vs_customers.xlsx`**.